

MSc Data Mining

Module outline: from raw data to a model you can defend

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1 • Aims

Turn raw, messy data into patterns a business can trust

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What students can do by the end

- Frame a business question as a data-mining task, and choose the right method for it
- Build and tune both **supervised** and **unsupervised** models
- Evaluate honestly: tell a real result from one that has fooled itself
- Reason about scale, data quality, and the ethics of acting on a model

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The arc of the module

- 1. Data preparation and exploration:** cleaning, integration, transformation; exploratory analysis; feature engineering; the data-leakage trap
- 2. Frequent patterns and association rules:** Apriori, FP-growth; support, confidence, lift; market-basket analysis
- 3. Classification:** decision trees, Naïve Bayes, k-NN, SVM, and ensembles (random forests, boosting)
- 4. Clustering:** k-means, hierarchical, DBSCAN; choosing k; cluster validity
- 5. Evaluation and validation:** train/validation/test, cross-validation, ROC and AUC, precision/recall, overfitting
- 6. Scale and special topics:** dimensionality reduction (PCA), anomaly detection, text mining, mining data streams, ethics

6 units

one module, six units: from data preparation to honest evaluation and scale

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Tools and datasets

- **Python** (pandas, scikit-learn) and **R** for the hands-on labs
- Open, real-world datasets (UCI and similar) rather than toy examples
- Every result is reproduced from committed code; any number traces back to the analysis behind it

A worked Python example for this module is public: `github.com/mtusman-ai/heart-failure-clinical-ml-py`.

How it is taught

- Lectures pair each method with the problem it was invented to solve
- Hands-on labs build one end-to-end mining pipeline, dataset to evaluated model
- The recurring discipline: validate before you believe, and name the leakage before it flatters the score

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Why it matters

A model is only as trustworthy as the evaluation behind it. This module teaches students to earn that trust before they ship a result.

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In summary

Core texts: Han, Kamber & Pei, Data Mining: Concepts and Techniques · James et al., An Introduction to Statistical Learning · Leskovec, Rajaraman & Ullman, Mining of Massive Datasets
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